

Preclinical public health modules

**Module title: Measurement of Health and Diseases
(Epidemiology and Biostatistics)**

Module Number COMH1012

ECTS = 10 1.75 ECTS = 1 credit hours = 16 hours

Placement: Preclinical Year I, Semester II

Weeks = 5

Pre-requisite module: Basic Concepts of Disease and Therapy

Module Description:

The module covers definitions, principles and concepts, estimating the occurrence of disease, interpreting the occurrence of disease by person, place, and time; validity and reliability and concepts of disease causation and determinants. Students are also introduced to the design and conduct of epidemiologic studies. This module also introduces theoretical and practical aspects of introductory biostatistics which is relevant to undergraduate medical training program.

Module Objective

At the end of this module, learners will be able to apply public health methods for the measurement of health and disease at population level.

Supporting Objectives

- Explain the notion of health from scientific and layman perspective
- Describe the history, evolution and functions of public health and epidemiology; its relevance to the practice of medicine
- Describe epidemiological approach to disease causation with emphasis on infectious diseases
- Describe levels of prevention regarding avoidance and control at different levels
- Describe types of epidemiologic studies including their use and limitations
- Calculate and interpret measures of morbidity and mortality including from existing data sources
- Measures of association (Odds ratio, relative risk)
- Describe criteria for establishing and evaluating screening programs and factors that affect validity and reliability of screening tests

- Describe the processes, uses, and evaluation of public health surveillance
- Apply the steps of an outbreak investigation and management
- Discuss epidemiology of diseases of public health significance in Ethiopia and locally
- Suggest health promotion and disease prevention methods for major public health problems
- Explain the basic principles of ethics in epidemiologic research
- Apply basic biostatistics concepts, tools and methods
- Understand data, types of data and scales of measurement
- Organize, summarize and present data
- Understand the concepts of probability, probability distributions
- Inferential statistics; sampling and sampling distributions
- Use information technology to manage information, access online medical information, and support one's own education
- Search, collect, organize and interpret health and health-related information from different sources

Teaching-Learning Methods

- Interactive lecture and discussion
- Small group learning activities: assignment, exercise, case study
- Individual reading
- Use of computer applications and access to the internet
- **Student presentations**

Teaching-Learning Materials

- AV aids (LCD and computer or Overhead projector and transparencies, writing board and marker or chalk)
- Computers with appropriate statistical software like EPI info and SPSS
- Handouts of lecture materials

Reference Books

1. Fletcher. Principles of Epidemiology
2. Charles H Hennekens and Julie E Buring. Epidemiology in Medicine
3. Rothman, Kenneth J.; Greenland, Sander; Lash, Timothy L. Modern epidemiology. 3rd edition. 2008
4. David G. Kleinbaum, Kevin M. Sullivan. A pocket guide to epidemiology. 2007
5. Yemane Berhane, Damen Hailemariam and Helmut Kloos. Epidemiology and ecology of health and disease in Ethiopia. 2006
6. Mausner and Bahn; Introductory Epidemiology
7. Daniel. Biostatistics: a foundation for analysis in health sciences.

8. Pagano. Principles of Biostatistics
9. Colton. Statistics in Medicine
10. Bland. An introduction to Medical Statistics.

Assessment Methods

- **Group assignment (school surveys) & presentation (30%)**
 - Data sets, filed epi case studies, prevalence assignments, using data sets like cancer registry etc...
- **Class attendance & participation (10%)**
- **Written exam (60%)**

Module Schedule

Week	Contents	Time/week
Week 1	1. Introduction to Public Health • Health and disease: concepts, definitions and perspectives • Public health: definition, philosophy, history, development, core functions and services • Public health sciences, their scope and use in medicine 2. General Principles of Epidemiology • History, definition, uses, scope, categories and assumptions of Epidemiology • Epidemiological concepts of disease causation • Epidemiological models in disease causation (epidemiological triangle, web of causation, wheel model) • Establishing causation and factors in causation 3. Concepts in Infectious Disease Epidemiology • Infectious process, chain of infection and mechanism of disease transmission • Transmission probability and estimation of transmission probability (Secondary attack rate & Binomial model) • Time lines of infection and Natural history of diseases (communicable and non- communicable)	◦ 20 contact hours

	• Levels of Disease prevention; Levels of Disease occurrence • Infection and Disease Outcome (Infectiousness, Pathogenicity and Virulence)	
Week 2	4. Description of the occurrence of disease in Human Population • Analysis by time, place and person • Epidemic or Outbreak: definition, types, epidemic curve 5. Measurement in Epidemiology • Epidemiologic variables (Exposure vs. Outcome) • Basic Measurements in Epidemiology (Count, ratio, proportion and rate) • Morbidity and Mortality measurement (Prevalence and Incidence) • Comparing disease occurrence (absolute and relative comparisons) • Measures of Association 6. Epidemiologic Study Designs • Descriptive studies; case reports, case series, correlational and cross-sectional studies • Analytical Studies (Observational and Interventional); case control, cohort and interventional studies 7. Evaluation of Evidence • Accuracy of Measurement • Bias, Confounding and Chance • Establishing causal association	▪ 20 contact hours

Week 3	8. Diagnostic and Screening tests • Definition and types of screening; criteria for screening • Factors affecting validity and reliability of screening tests 9. Using available information to measure health and disease (health information system) 1. Epidemiology of diseases of public health significance in Ethiopia • Epidemiology of ARI, VPD, TB, STI and HIV • Epidemiology of Vector Borne Disease • Epidemiology of Zoonotic diseases 2. Public health surveillance • Principles of public health surveillance • Integrated disease surveillance and response • Timely warning and intervention 3. Outbreak investigation and management • Patterns of occurrence of diseases • Disease outbreaks • Steps of investigation of an outbreak • Management and control of an outbreak or epidemic 4. Basic principles of ethics in epidemiologic research	■ 20 contact hours
Week 4	5. Introduction to Biostatistics 6. Data and types of data 7. Scales of Measurement 8. Presenting and summarizing data • Frequency distribution, measures of central tendency and measures of dispersion • Frequency distribution 9. Probability and probability distributions 10. Sampling and Sampling distributions	■ 20 contact hours

Week 5	11. Statistical inferences • Point and interval estimation • Hypothesis testing • Test of significance • Concept of P-value • Hypothesis testing 12. Sample size determination 13. Data summarization and presentation	▪ 20 contact hours